DATA VISUALIZATION & PREDICTIVE ENGINEERING

Credit Risk & Loan Default Predictive Pipeline

An academic defense of algorithmic risk assessment, data preprocessing, and production deployment.

Project Objective & Problem Statement

- **Core Financial Risks:** Mitigation of unrecoverable capital allocation.
- **Non-Performing Assets (NPAs):** Early detection pipeline to prevent portfolio degradation.
- **Lifecycle Scope:** From raw applicant data ingestion to cloud-based prediction endpoints.
- **Strategic Imperative:** Transitioning from reactive collection to proactive underwriting.

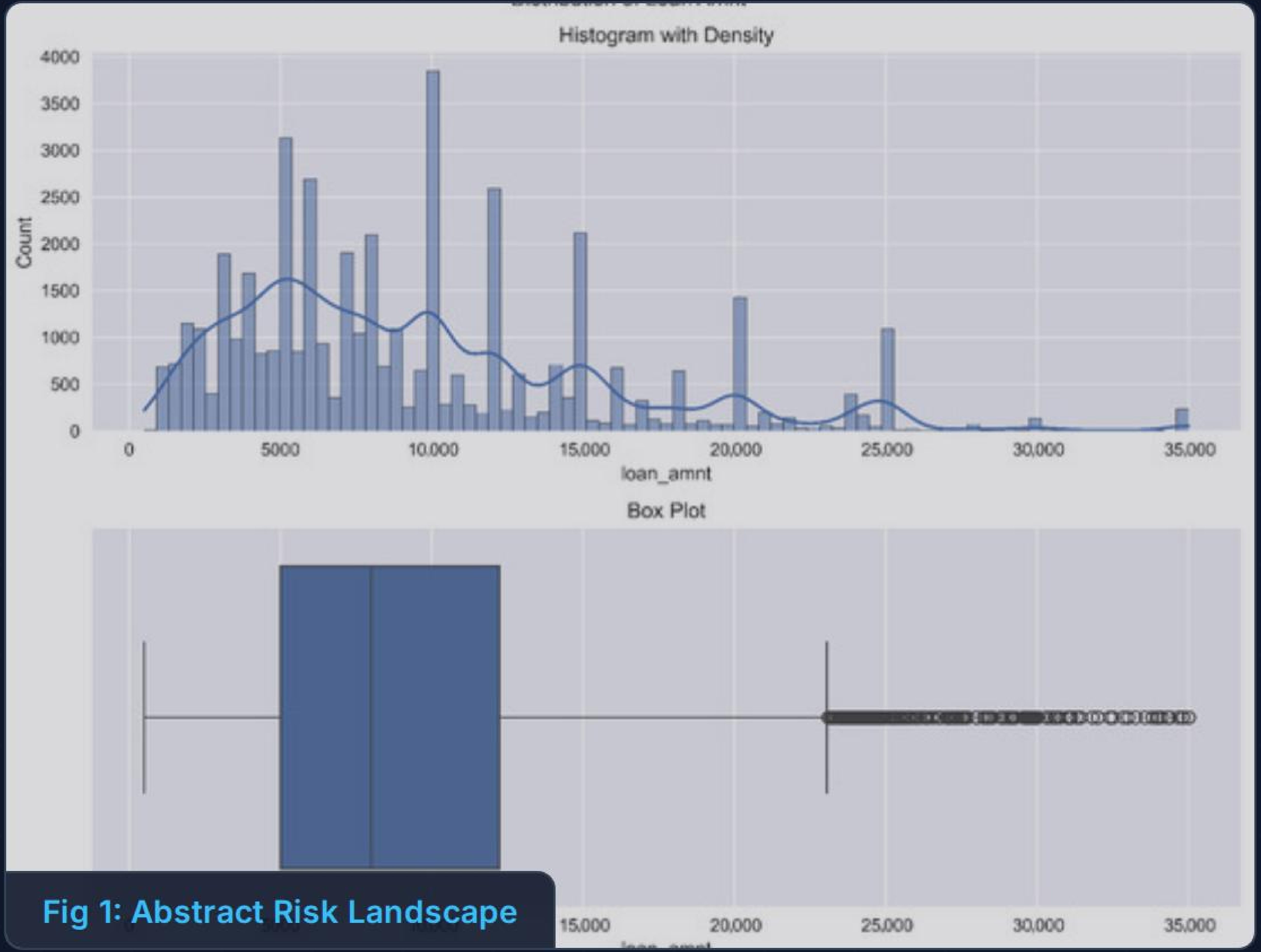


Fig 1: Abstract Risk Landscape

SPEAKER NARRATIVE

"Welcome. Today we present our predictive pipeline designed to tackle one of banking's most critical vulnerabilities: loan defaults. By predicting NPAs before capital is deployed, we shift the risk lifecycle from reactive loss-mitigation to proactive intelligence."

Data Preprocessing Pipeline

Identifier Drop

Removal of high-cardinality noise, specifically dropping the LoanID column to prevent over-indexing on structural artifacts.

Statistical Imputation

Executing Median imputation for continuous variables and Mode imputation for categorical voids to preserve distribution integrity.

Row Deduplication

Purging identical records to eliminate synthetic weighting and prevent data leakage during model training.

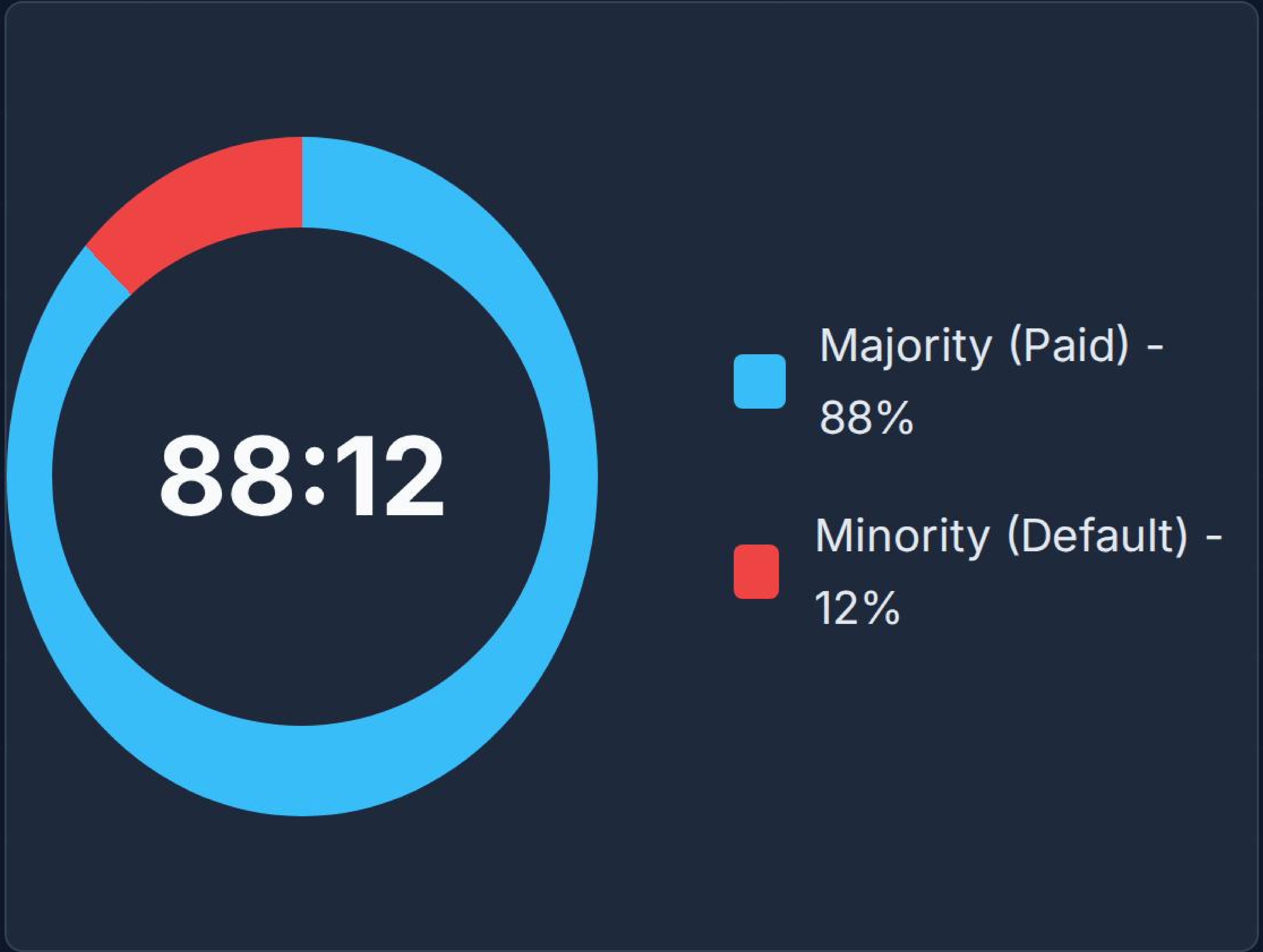
IQR Outlier Clipping

Applying Interquartile Range thresholds to clip extreme values, stabilizing the model against anomalous financial leverage inputs.

SPEAKER NARRATIVE

"Raw financial data is inherently noisy. Our pipeline strips out synthetic noise like LoanIDs, intelligently imputes missing values using medians, drops duplicates to prevent data leakage, and clips severe outliers using standard IQR boundaries to ensure model stability."

Target Class Imbalance

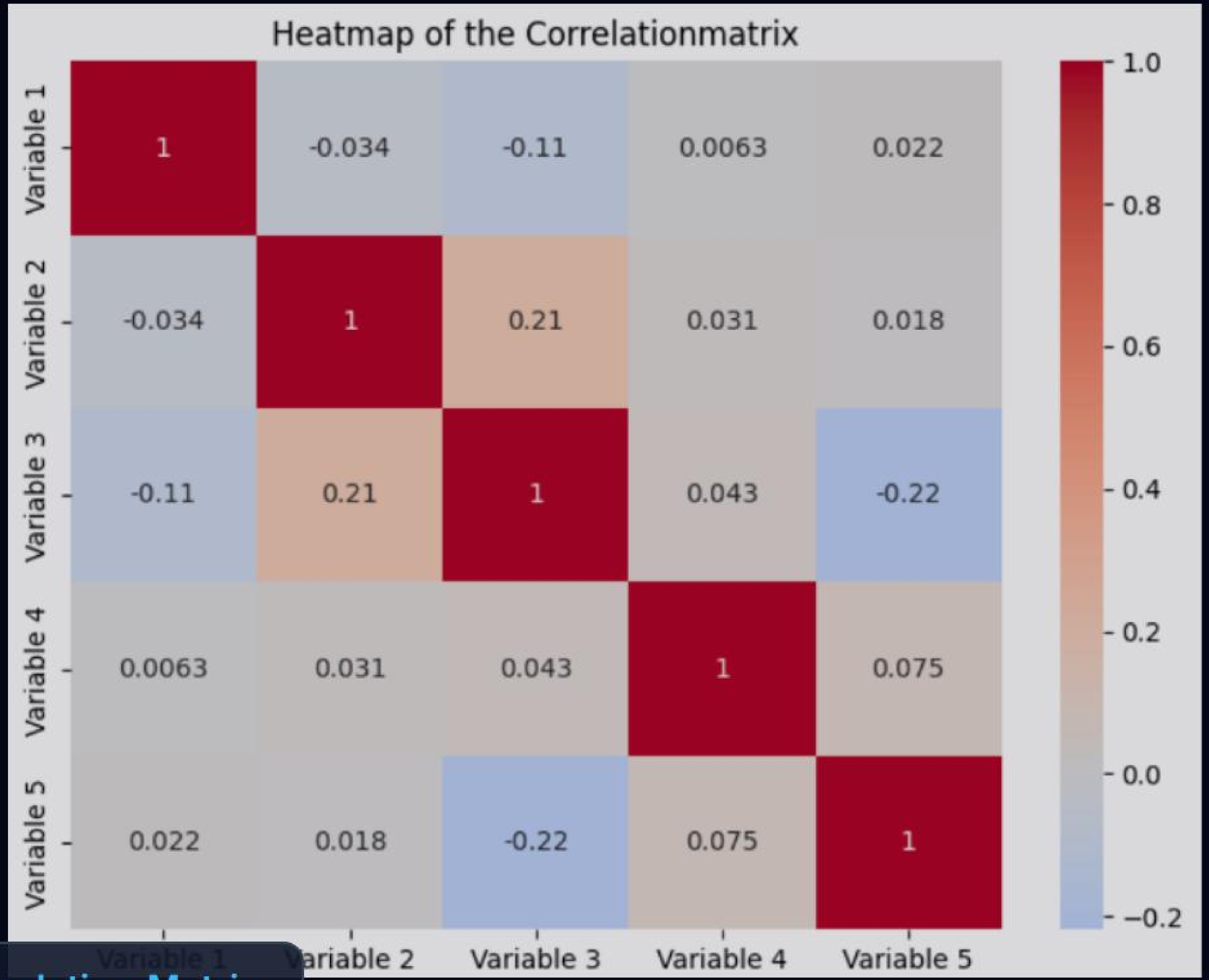


- **Distribution Skew:** Heavy weighting toward successful repayment.
- **Diagnostic Risk:** High accuracy can be achieved by simply predicting '0' (Paid) universally.
- **Metric Shift:** Necessitates evaluating F1-Score and Recall over raw Accuracy.
- **Algorithmic Challenge:** The model naturally starves for default pattern recognition.

> **SPEAKER NARRATIVE**
>
> *"Visualizing our target variable reveals a severe 88 to 12 imbalance. While typical for financial datasets, this presents a diagnostic risk; a naive model could achieve 88% accuracy simply by denying that defaults exist. This mandates a shift away from accuracy as our primary metric."*

Feature Interactions & Independence

- **Independence Verified:** Correlation matrix confirms near-zero multi-collinearity between primary predictors.
- **Distinct Correlates:** Features offer unique, non-overlapping signals for predictive modeling.
- **Dimensionality:** No PCA (Principal Component Analysis) required due to orthogonal feature behavior.

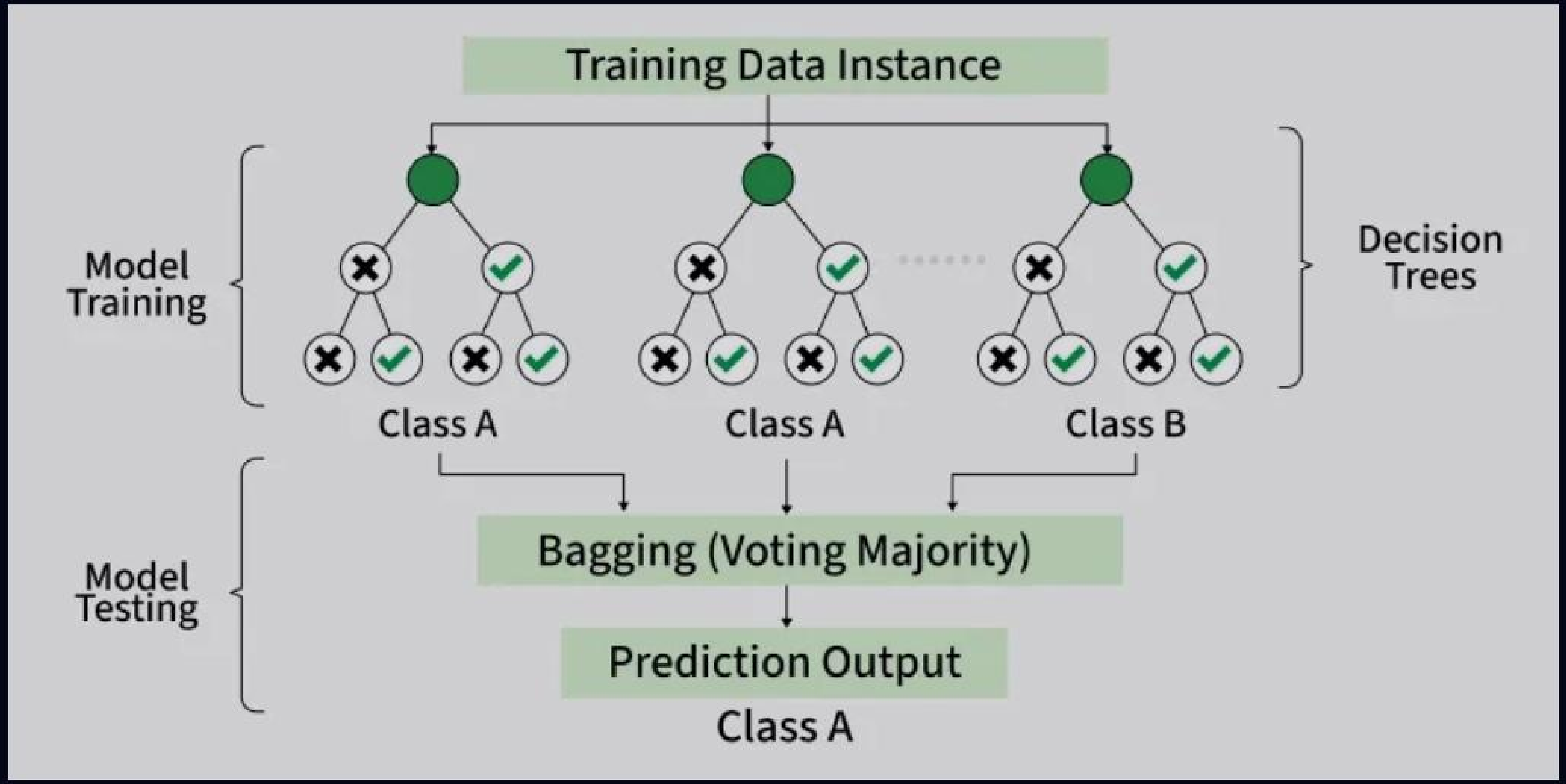


Correlation Matrix

SPEAKER NARRATIVE

"Referencing the correlation matrix, we validate feature independence. The heatmap confirms minimal multi-collinearity among inputs. Because our variables are orthogonal, they each contribute distinct predictive power without the need for dimensionality reduction."

Model Architecture & Validation



Architecture Diagram

- **Algorithm Selection:** Random Forest Classifier chosen for non-linear boundary handling.
- **Hyperparameters:** Configured with `n_estimators=100` and `max_depth=12` to prevent overfitting.
- **Validation Strategy:** 80/20 Stratified Train-Test Holdout method.
- **Ensemble Advantage:** Bagging mechanism reduces variance across the skewed dataset.

SPEAKER NARRATIVE

"For our architecture, we deployed a Random Forest ensemble utilizing 100 estimators capped at a depth of 12. We validated this using a strict 80/20 stratified holdout to ensure the train and test sets perfectly mirrored our highly imbalanced class distribution."

The Baseline Breakdown

```
==================== MODEL PERFORMANCE REPORT =====================

Overall Accuracy Score: 0.8859

Classification Matrix:
                precision    recall  f1-score   support

 Fully Paid (0)       0.89      1.00      0.94     45139
  Defaulted (1)       0.69      0.03      0.06      5931

       accuracy                           0.89     51070
      macro avg       0.79      0.52      0.50     51070
   weighted avg       0.86      0.89      0.84     51070

===================================================================
```

- **The Accuracy Trap:** Baseline model achieved an impressive-sounding 88.59% overall accuracy.
- **The Reality:** As shown in the classification report, it successfully identified only 3% of actual defaulting accounts (Recall = 0.03).
- **Business Impact:** This unacceptable false-negative rate exposes the institution to massive unrecognized capital risk.

SPEAKER NARRATIVE

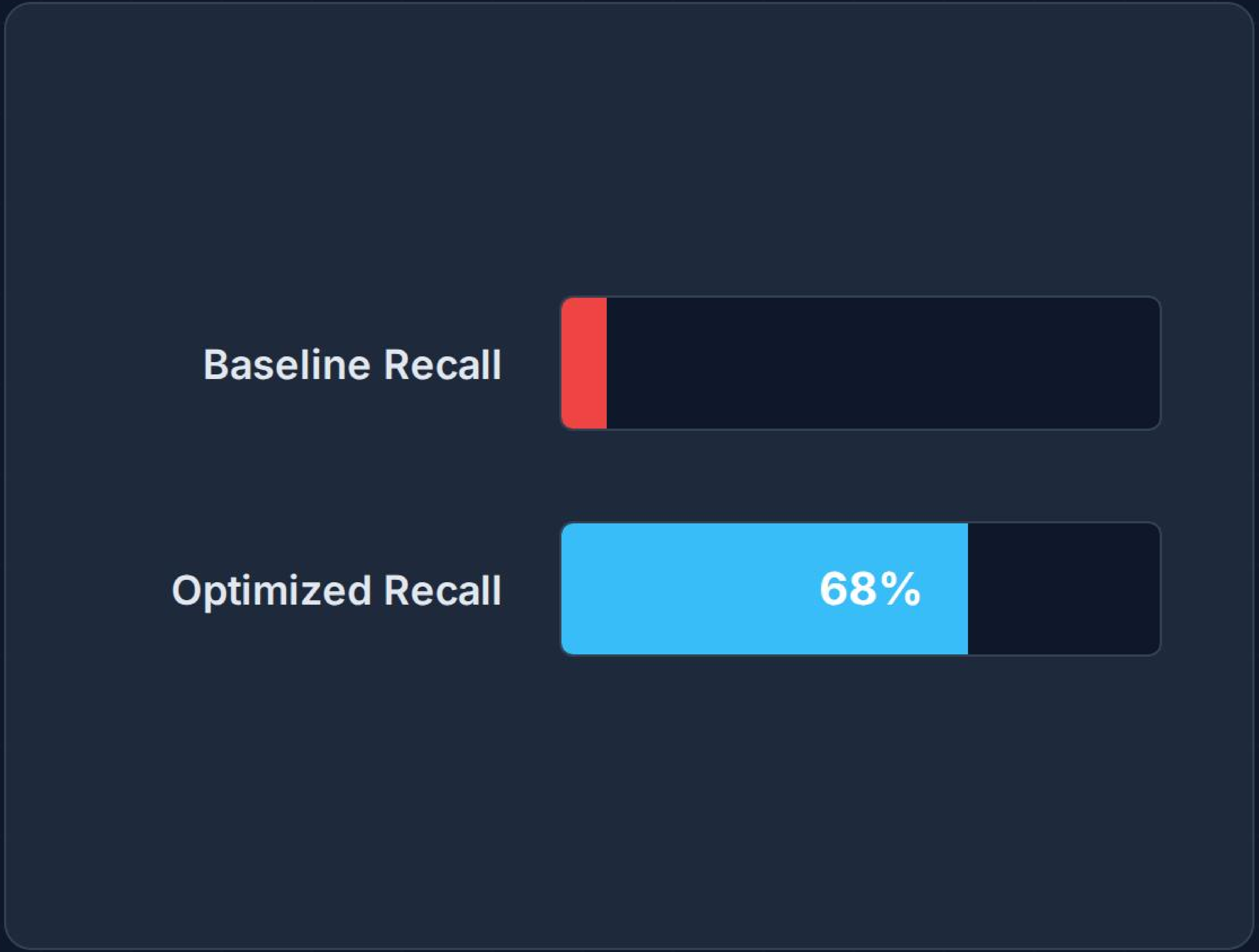
"Here we see the 'Accuracy Trap' in action. Out of the box, the model hit 88.59% accuracy. However, looking at the classification matrix, the recall for the defaulted class is a dismal 0.03, capturing only 3% of actual defaults. In a production environment, this translates to massive, undetected capital exposure."

Optimization via Class Balancing

- **Parameter Injection:** Implementing `class_weight='balanced'` within the estimator.
- **Algorithmic Shift:** Penalizes the model heavily for missing the minority default class.
- **Threshold Adjustment:** Shifts the internal decision boundaries to catch high-risk accounts aggressively.
- **Trade-off:** Willingly sacrificing minor precision on good loans to vastly improve default recall.

SPEAKER NARRATIVE

"To fix the recall collapse, we injected the 'balanced' class weight parameter. This mathematically penalizes the model for missing defaults, effectively shifting decision thresholds. We trade away a small fraction of overall accuracy to achieve a massive operational gain in identifying true risks."



Actionable Operational Insights

Age-Based Risk

Data indicates specific demographic cohorts present higher instability, allowing for targeted loan product routing.

Interest Rate Thresholds

Identified distinct tipping points where high interest rates shift borrowers from stable to default-prone statuses.

Leverage Interaction

The Debt-to-Income (DTI) density profiles reveal severe risk scaling when high leverage meets low tenure.

SPEAKER NARRATIVE

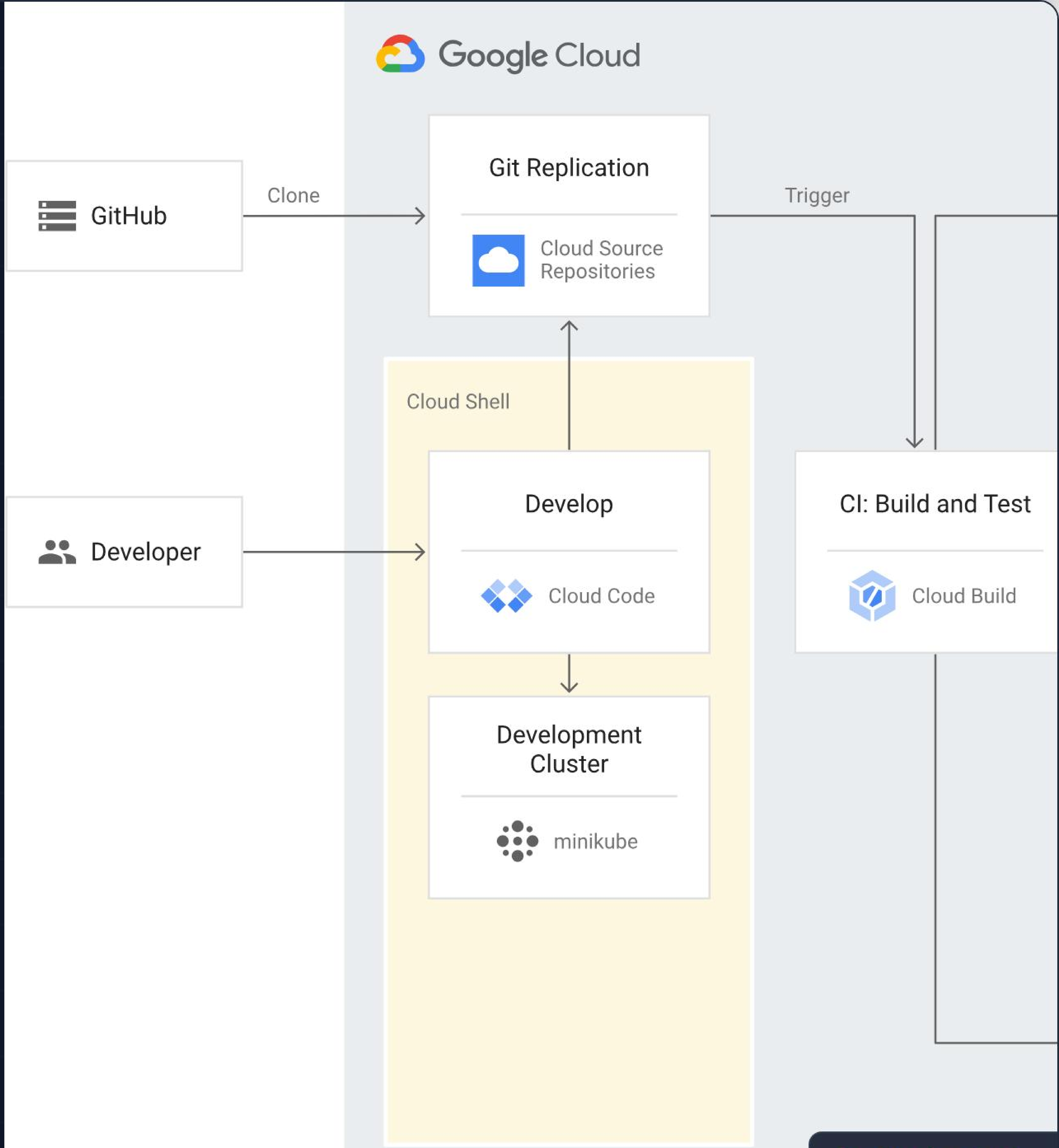
"Beyond just predicting a 1 or 0, the model yields operational intelligence. We've mapped specific age-based risk profiles, identified the exact interest rate thresholds that trigger defaults, and mapped the compounding interaction between high Debt-to-Income ratios and low employment tenure."

Interactive UI & Cloud Production

- **Streamlit Architecture:** Front-end application wrapper enabling real-time risk scoring for loan officers.
- **Memory Optimization:** Data processing capped at 40,000 rows to successfully bypass the rigid 1GB RAM limits of cloud hosting instances.
- **Repository Structure:** Containerized approach combining standard serialized `.pkl` models with stateless inference endpoints.

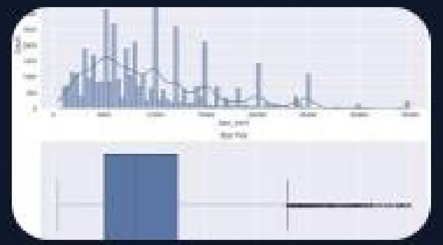
SPEAKER NARRATIVE

"Finally, we bridged the gap between local notebook and production. We wrapped the model in a Streamlit architecture for interactive UI. Crucially, we engineered memory optimizations—sampling at 40,000 rows—to ensure seamless execution within strict 1GB cloud server limits. The repository is live."



Production Arch.

Image Sources



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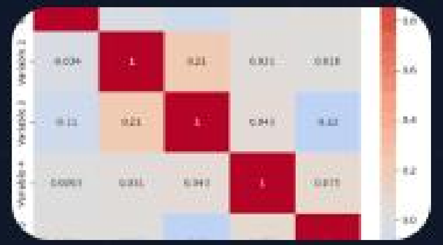

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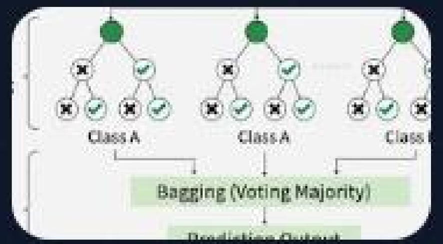

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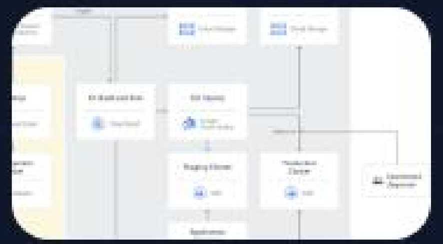

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